Division in $\mathbb{Q}$

1. $\frac{4}{5} \div \frac{2}{3}$
2. $\frac{7}{9} \div \frac{14}{15}$
3. $-\frac{5}{8} \div \frac{15}{16}$
4. $\left(-\frac{9}{10}\right) \div \left(-\frac{3}{5}\right)$
5. $3\frac{1}{3} \div 1\frac{1}{4}$
6. $-4\frac{1}{2} \div 2\frac{1}{4}$
7. $\left(-2\frac{2}{5}\right) \div \left(-1\frac{1}{15}\right)$
8. $\dfrac{-\frac{5}{6}}{\frac{10}{9}}$
9. A ribbon of length $\frac{7}{8}$ meters is cut into equal pieces, each measuring $\frac{1}{16}$ meters long. How many pieces of ribbon are obtained?
10. A container holding $5\frac{1}{4}$ liters of juice is poured equally into smaller bottles, each with a capacity of $\frac{3}{4}$ liters. How many full bottles can be filled?

Solutions

1. $\frac{6}{5}$ **or** $1\frac{1}{5}$

   Explanation: Multiply by reciprocal of divisor:

   $$\frac{4}{5} \div \frac{2}{3} = \frac{4}{5} \times \frac{3}{2} = \frac{12}{10} = \frac{6}{5} = 1\frac{1}{5}.$$

2. $\frac{5}{6}$

   Explanation: Multiply by reciprocal and cancel factors:

   $$\frac{7}{9} \div \frac{14}{15} = \frac{7}{9} \times \frac{15}{14} = \frac{1 \times 5}{3 \times 2} = \frac{5}{6}.$$

3. $-\frac{2}{3}$

   Explanation: Multiply by reciprocal:

   $$-\frac{5}{8} \div \frac{15}{16} = -\left(\frac{5}{8} \times \frac{16}{15}\right) = -\left(\frac{1}{1} \times \frac{2}{3}\right) = -\frac{2}{3}.$$

4. $\frac{3}{2}$ **or** $1\frac{1}{2}$

   Explanation: Quotient of two negative numbers is positive:

   $$\left(-\frac{9}{10}\right) \div \left(-\frac{3}{5}\right) = \frac{9}{10} \times \frac{5}{3} = \frac{3 \times 1}{2 \times 1} = \frac{3}{2} = 1\frac{1}{2}.$$

5. $\frac{8}{3}$ **or** $2\frac{2}{3}$

   Explanation: Convert mixed numbers to improper fractions:

   $$3\frac{1}{3} = \frac{10}{3}, \quad 1\frac{1}{4} = \frac{5}{4}.$$

   Multiply by reciprocal:

   $$\frac{10}{3} \div \frac{5}{4} = \frac{10}{3} \times \frac{4}{5} = \frac{2 \times 4}{3 \times 1} = \frac{8}{3} = 2\frac{2}{3}.$$

6. -2

   Explanation: Convert to improper fractions:

   $$-4\frac{1}{2} = -\frac{9}{2}, \quad 2\frac{1}{4} = \frac{9}{4}.$$

   Multiply by reciprocal:

   $$-\frac{9}{2} \div \frac{9}{4} = -\left(\frac{9}{2} \times \frac{4}{9}\right) = -\left(\frac{1}{1} \times \frac{2}{1}\right) = -2.$$

7. $\frac{9}{4}$ **or** $2\frac{1}{4}$

   Explanation: Convert to improper fractions:

$$-2\frac{2}{5} = -\frac{12}{5}, \quad -1\frac{1}{15} = -\frac{16}{15}.$$

   Multiply by reciprocal:

$$\left(-\frac{12}{5}\right) \times \left(-\frac{15}{16}\right) = \frac{3 \times 3}{1 \times 4} = \frac{9}{4} = 2\frac{1}{4}.$$

8. $-\frac{3}{4}$

   Explanation: Complex fraction represents division $\left(-\frac{5}{6}\right) \div \left(\frac{10}{9}\right)$:

$$-\frac{5}{6} \times \frac{9}{10} = -\left(\frac{1 \times 3}{2 \times 2}\right) = -\frac{3}{4}.$$

9. 14 **pieces**

   Explanation: Divide total length by piece length:

$$\frac{7}{8} \div \frac{1}{16} = \frac{7}{8} \times 16 = 7 \times 2 = 14 \text{ pieces}.$$

10. 7 **bottles**

    Explanation: Convert total volume $5\frac{1}{4} = \frac{21}{4}$ and divide:

$$\frac{21}{4} \div \frac{3}{4} = \frac{21}{4} \times \frac{4}{3} = \frac{21}{3} = 7 \text{ bottles}.$$